

Modeling

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Practice quiz

In order to participate in a cooking competition called "Top Chef", Frank must build a team of excellent chefs. He has access to a pool of candidates (see Table 1). For each of them, he knows the age, the restaurant where they come from, and their cooking specialties (some candidates have only one, some others have two). He also knows the number of points that each of them has been able to collect in previous similar competitions. And he wants to build a team with the maximum number of such points. But several constraints apply:

1. The team must contain at least 3 chefs who are skilled in preparing appetizers.
2. The team must contain at least 4 chefs who are skilled in preparing fish.
3. The team must contain at least 4 chefs who are skilled in preparing meat.
4. The team must contain at least 3 chefs who are skilled in preparing dessert.
5. To promote young people, the team must contain at least 2 chefs who are 20 or less.
6. Yoda and Obi-Wan cannot stand each other, and it is not possible to have both of them in the team.
7. Han and Sheev have so much experience in working together that if one of the two is included in the team, the other must be too.

Name	Age	Restaurant	Points	First specialty	Second specialty
Anakin	49	Lausanne	124	Fish	Meat
Chewbacca	39	Noirmont	105	Fish	Meat
Dooku	49	Crissier	118	Meat	Dessert
Gial	33	Mézières	134	Dessert	
Han	35	Crissier	160	Appetizer	
Jabba	47	Crissier	184	Fish	
Jubnuk	28	Neuchatel	101	Fish	
Lando	17	Lucens	116	Dessert	
Leia	18	Sierre	187	Appetizer	Fish
Lorth	41	Baulmes	132	Dessert	
Luke	37	Vufflens	181	Appetizer	
Obi-Wan	22	Vufflens	199	Fish	Meat
Padme	38	Cossonay	192	Appetizer	
Qui-Gon	19	Cossonay	136	Fish	
Sebulba	43	Valeyres	123	Meat	
Sheev	21	Crissier	127	Meat	
Teebo	32	Sierre	132	Meat	
Watto	20	Sierre	131	Meat	
Yoda	39	Noirmont	102	Appetizer	Fish
Zuckuss	42	Vufflens	123	Dessert	

Table 1: Data about each chef

8. For the sake of fairness, not more than 3 chefs from the same restaurant should be hired in the same team.

Write an integer optimization problem to help Frank build his team.